AI Tool Helps Scientists Read the Messages Cells Send Each Other

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Every living body starts as a small cluster of **stem cells**. These "blank slate" cells slowly become specialized, turning into brain, liver or muscle cells. What each cell becomes depends on which of its **genes** are switched on or off.

Cells do not decide alone. They constantly send and receive chemical signals with their neighbors. These messages travel through step-by-step routes called **signaling pathways**, which change which **genes** are active. Scientists had assumed each pathway worked differently in each type of cell, so mapping them all would take a very long time.

Now a team led by MIT biology professor Pulin Li and graduate student Nicholas Hutchins has found that each signaling pathway leaves a unique "fingerprint," a pattern of gene activity. The same pathway leaves a similar fingerprint in many types of cells.

The team built an **artificial intelligence** model called IRIS to spot these fingerprints. They trained it on data showing how thousands of human **embryonic stem cells** responded to combinations of six major **signaling pathways**. IRIS can then estimate which signals a cell received, even in types of cells it has never seen before.

"Think of voice recognition systems like Siri, which are trained mainly in English, but then use that training to help them recognize other languages," Li said.

When tested on cells from mouse embryos, IRIS correctly predicted signals in cells that would become heart, gut, muscle and spinal cord tissue. It also predicted a signal that would help create a cell type important for lung development, and experiments in mouse embryos confirmed it.

The findings were published in the journal Nature Methods. They could help scientists grow better lab models of organs to study diseases and test treatments.

Word Treasure

stem cells -- Stem cells are basic cells that can develop into many different kinds of cells in the body.

genes -- Genes are instructions inside cells that help decide how a living thing grows and works.

signaling pathways -- Signaling pathways are chains of steps that carry a chemical message into a cell and change what it does.

embryonic -- Embryonic describes something from an embryo, the very early stage of a developing animal or person.

artificial intelligence -- Artificial intelligence is computer technology that learns patterns from data to make predictions or decisions.

Background you need

Every cell in a body carries the same genes, but different cells switch on different genes, which is why a muscle cell is different from a brain cell.

As an embryo develops, cells send chemical signals to their neighbors that help guide what each cell becomes.

Machine learning is a kind of artificial intelligence that finds patterns in large amounts of data and uses them to make predictions.

Scientists also grow organoids, tiny lab models of organs, and knowing the right signals helps them make these models more accurately.

Break it down

LEAD: Stem cells become specialized partly through chemical signals from neighbors
PROBLEM: mapping every pathway in every cell type would take a very long time
KEY DISCOVERY: each pathway leaves a consistent gene-activity fingerprint
+- WHO: MIT's Pulin Li and graduate student Nicholas Hutchins
+- TOOL: the AI model IRIS, trained on thousands of human embryonic stem cells
EVIDENCE: correct predictions in mouse embryo cells for heart, gut, muscle and spinal cord; a lung prediction confirmed
PUBLISHED: journal Nature Methods
NEXT: better lab-grown organ models to study diseases and test treatments

Why it matters

Knowing which signals turn stem cells into specific cells could help scientists grow better models of organs in the lab. Those models can be used to study diseases and test new treatments.

Test what you remember

1. What decides what type of cell a stem cell becomes?

- (A) Which of its genes are switched on or off
- (B) The size of the embryo
- (C) The color of the cell
- (D) The temperature outside

2. What did the researchers discover about each signaling pathway?

- (A) It cannot be measured
- (B) It changes every day
- (C) It only works in brain cells
- (D) It leaves a unique fingerprint of gene activity

3. What is the name of the AI model?

- (A) Dragon
- (B) IRIS
- (C) MIA
- (D) Siri

4. Why did Pulin Li compare IRIS to Siri?

- (A) Both can talk out loud
- (B) Both use training in one area to recognize new ones
- (C) Both only work in English
- (D) Both were made by MIT

5. How many major signaling pathways were in the training data?

- (A) Thirteen
- (B) A thousand
- (C) Six
- (D) Two

6. What prediction did experiments in mouse embryos confirm?

- (A) A signal that helps create a cell type important for lung development
- (B) A new type of muscle
- (C) A way to change eye color
- (D) A cure for all diseases

Answer key (try the quiz first!): 1: A 2: D 3: B 4: B 5: C 6: A

Questions to think about

MY THOUGHTS (at least 20 words):